

Realistic Impact Perception in Multi-User VR Tennis: Enhancing UX through Hit-Stop and Modification of Others' Movement

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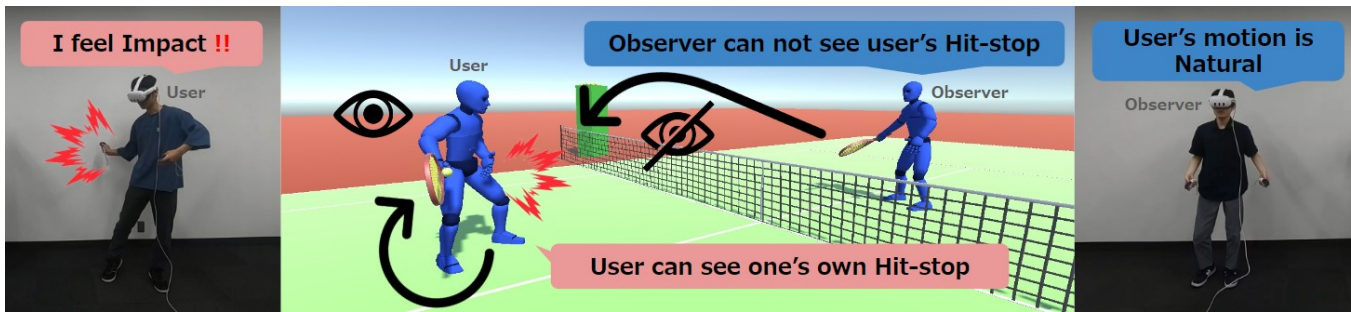


Figure 1: Hit-stop asymmetric presentation method: the left player can see their own hit-stop and feel a pseudo-impact force sense, while the right player can't see the left player's hit-stop and is presented with the motion converted to a natural one.

Abstract

Hit-stop is an expression technique mainly used in fighting games to emphasize an impact response by slowing or stopping a character's motion. A previous study also found that it can provide a pseudo-impact virtual reality (VR) sensation [Ban and Ujitoko 2021]. However, in VR, observers of a hit-stop often feel uncomfortable when they see the unpredictable movements of the hit-stopping player. In this study, we propose a hit-stop asymmetric presentation method to apply a hit-stop to multi-user VR. We design a hit-stop presentation system in which hit-stops are applied to the user's own motion, while the observer is presented with the user's motion converted to a neutral one without hit-stops. Our method could significantly reduced discomfort caused by an opponent's hit-stopping motion while maintaining a player's own pseudo-impact sensation for hit-stop [Noguchi et al. 2024]. In our demonstration, we apply this method to a VR tennis environment where two players can play simultaneously. The users can feel that their opponent's motion is natural, while they perceive a pseudo-impact force sense with their own hit-stops. Therefore, the proposed method should be useful for improving the immersive experience of players in multi-user VR.

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SA Emerging Technologies '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1137-4/24/12

<https://doi.org/10.1145/3681755.3688941>

CCS Concepts

• Human-centered computing → Haptic devices; Virtual reality.

Keywords

hit-stop, pseudo-haptics, discomfort, multi-user virtual reality

ACM Reference Format:

Shinnosuke Noguchi, Keigo Matsumoto, Yuki Ban, and Takuji Narumi. 2024. Realistic Impact Perception in Multi-User VR Tennis: Enhancing UX through Hit-Stop and Modification of Others' Movement. In *SIGGRAPH Asia 2024 Emerging Technologies (SA Emerging Technologies '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681755.3688941>

1 Introduction

Pseudo-haptics is a method of perceiving tactile and/or force sensations without requiring a mechanical tactile interface [Lécuyer 2009]. In this phenomenon, users perceive tactile and/or force sensations when they observe the displacement between their real and virtual bodies [Ujitoko and Ban 2021].

Hit-stop is a type of pseudo-haptics in the broad sense of the term and is a method for making people feel the force of impact using visual tricks [Ban and Ujitoko 2021]. This technique emphasizes the response to an attack by stopping the movement of both the attacker and the attacked for a short period (less than one second). It is mainly used in fighting games such as Street Fighter ¹ and Super

¹<https://www.streetfighter.com/6/ja-jp>

Smash Bros². Hit-stop applied to virtual environments presents the user with the perception of impact force and an extremely fresh experience for the user, resulting in a more immersive experience in VR without additional haptic equipment.

Although hit-stop could be an easy method for presenting impact force perception, it causes a sense of discomfort when the user observes another person with hit-stops in a virtual space shared by many people (multi-user VR) [Noguchi et al. 2024]. In multi-user VR, users observe the detailed movements of avatars that represent their real bodies and communicate with them in a manner similar to reality. In such interactions with avatars, sudden changes in the avatar's movements that would not occur in reality may cause confusion among observers. In this study, we propose a hit-stop asymmetric presentation method to reduce the discomfort of motion with hit-stops by others in multi-user VR. This approach compensates for the user's own motion with a hit-stop while their movements are displayed to the other player as uncorrected with a hit-stop. By preventing the observer from knowing that the user is presented with a hit-stop, the proposed method enhances the immersion of the experience of the observer in multi-user VR (Fig. 1).

2 System Design

In this study, we propose a novel hit-stop presentation method in which the user is presented with a hit-stop and feel a pseudo-impact sensation, while the other player is presented with the user's motion converted to a natural one (Fig. 1). This method allows the observer to continue playing without being aware of the hit-stop applied to the user's move. Our method was designed based on the hit-stop presentation method proposed by Ban et al. [Ban and Ujitoko 2021]. From the moment a hit-stop is initiated, the movement of the avatar is delayed from that of the controller; when the hit-stop ends, the avatar is made to catch up with the controller by speeding up (Fig. 2).

The differences from the previous study are the existence of an opponent's avatar with a hit-stop and the specification that a hit-stop is also applied to the ball in the same way as the avatar and racket. Therefore, in the proposed method, when the user is hit-stopping, the movement of the user's avatar from the other player's viewpoint perfectly matches the controller's movement without delay. Consequently, in this method, the images observed by the user and observer during the hit-stop differ.

We constructed a VR tennis environment to apply the proposed method where two people could play simultaneously using Unity 2021³. The size of the tennis court was set to 23.77 m x 10.97 m, and the tennis ball was 68 mm in diameter and 58 g in weight, similar to those in a real environment. The VE was implemented using Oculus Integration⁴, which is a plugin widely used in VR researches. In addition, Final IK⁵ and Mirror⁶ were used for the tracking of avatars to real bodies and synchronization of the objects. In this demonstration, two users face each other over a net with a racket and play a tennis rally.

²https://www.smashbros.com/ja_JP/

³<https://unity.com/>

⁴<https://assetstore.unity.com/packages/tools/integration/oculus-integration-deprecated-82022>

⁵<https://assetstore.unity.com/packages/tools/animation/final-ik-14290>

⁶<https://assetstore.unity.com/packages/tools/network/mirror-129321>

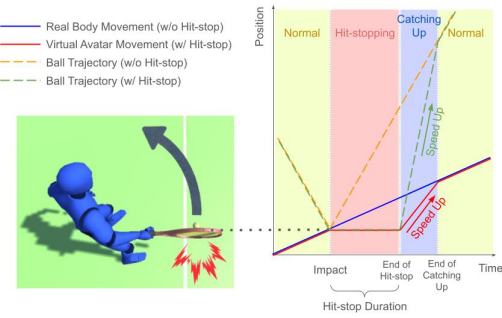


Figure 2: Relationship between the racket and the actual body movement

3 Findings

Our study [Noguchi et al. 2024] demonstrated that the discomfort with the opponent's motion was significantly reduced when normal motion without hit-stops was displayed using the hit-stop asymmetric presentation method, compared to when the opponent's motion was corrected by hit-stops. This suggests that the proposed method effectively reduces discomfort caused by an opponent's hit-stop. In addition, the perceived impact force was found to be greater under conditions in which one's own motion was halted than under the no hit-stop condition, which is consistent with the results of the study by Ban et al. [Ban and Ujitoko 2021]. Based on these results, the proposed method can improve the quality of the VR experience by making the user perceive the impact force on their own hit-stops while reducing the discomfort of others' hit-stops.

4 Demonstration

Our proposed method provides a fresh VR experience in multi-user VR by making the user perceive the impact force sensed by their hit-stop, while the other user's hit-stop is unnoticeable. At SIGGRAPH Asia 2024 Emerging Technologies, we will demonstrate VR tennis, which can be played by two people simultaneously, allowing them to experience the effect of hit-stops on their perception. In the system used for the experiment [Noguchi et al. 2024], the direction of the returned ball was fixed and hit back in one bounce, but at SIGGRAPH Asia, the system will be implemented so that the player can walk to some extent in real space and decide how and in which direction to hit the ball by him/herself.

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